Quantification: 542_z06
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	16
interior lacuna count	13
frame-edge lacuna count	3

Interior lacunae only: mean, median and sample SD (n = 13)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2541.77	3013	1427.44	px^2
major_axis_length_px	lacuna shape	108.237	124.464	40.2415	px
minor_axis_length_px	lacuna shape	31.7238	33.1799	10.2747	px
aspect_ratio	lacuna shape	3.5771	3.6285	1.3089	unitless
eccentricity	lacuna shape	0.935	0.9613	0.0596	unitless
solidity	lacuna shape	0.7472	0.7672	0.1322	unitless
roots_count	headline	10.9231	9	4.4993	count
ring_length_r30_px	headline	422.077	391	169.983	px
ring_length_r60_px	reported	1022.46	975	363.345	px
owned_length_px	ownership-dependent	2108.37	1773.18	1287.23	px
edge_count	ownership-dependent	88.1538	79	56.4341	count
mean_edge_length_px	ownership-dependent	24.5932	24.3648	3.2562	px
perimeter_px	normalised	297.812	309.522	115.117	px
ring_area_r30_px2	normalised	10771.7	11337	2967.43	px^2
ring_area_r60_px2	normalised	25565.5	27170	6351.05	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.9908	1	0.0272	unitless
ring_density_r30	normalised	0.0383038	0.0383699	0.00697074	px^-1
ring_density_r60	normalised	0.0393404	0.0389187	0.00589092	px^-1
roots_per_100px_perimeter	normalised	3.7216	3.7827	0.7614	per 100 px
ring_attached_length_r30_px	network v2	356.462	313	172.827	px
ring_attached_length_r60_px	network v2	745.846	668	398.961	px
ring_length_w_r30_px	network v2	478.799	441.161	195.496	px
ring_length_w_r60_px	network v2	1157.47	1100.96	416.848	px
sholl_crossings_r10	network v2	9.4615	9	4.1955	count
sholl_crossings_r20	network v2	13.4615	12	5.2697	count
sholl_crossings_r30	network v2	14.0769	13	5.5597	count
ring_width_mean_r30_px	width	6.6809	6.6928	0.4577	px

Per field
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measure	value	units
analysed_area_px2	1.00906e+06	px^2
total_skeleton_length_px	37752	px
canalicular_length_density_per_px	0.037413	px^-1
junction_count	748	count
junction_density_per_px2	0.00074128	px^-2
lacuna_count	16	count
lacunae_per_px2	1.586e-05	px^-2
skeleton_component_count	284	count

mean_component_length_px	132.93	px
median_component_length_px	27.5	px
field_length_density_w_per_px	0.0423479	px^-1
field_density_without_flagged_per_px	0.0373781	px^-1
field_density_in_roi_per_px		px^-1

Canalicular width, the new measure

measure	value	units
width_mean_px	7.3668	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10	px
width_pixels_used	36570	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	1381	54.5303	43.8441	1.2437	0.5946	0.657	-0.6304	16.7227	909.274	True
2	445	56.9837	11.5357	4.9398	0.9793	0.7672	-0.2356	148.654	554.791	False
3	556	44.867	25.9744	1.7274	0.8154	0.5265	-0.4071	66.9281	106.103	False
4	895	46.7108	27.3564	1.7075	0.8106	0.7956	0.2016	143.341	635.06	False
5	2450	125.59	27.3191	4.5971	0.9761	0.6618	-0.1995	179.65	380.673	False
6	4645	128.217	49.4579	2.5925	0.9226	0.72	-0.0979	201.844	870.357	False
7	2177	80.2406	35.2894	2.2738	0.8981	0.8959	-0.0663	304.003	624.629	False
8	3251	124.464	40.9538	3.0391	0.9443	0.639	0.028	423.426	66.5638	False
9	3239	125.767	33.4856	3.7559	0.9639	0.9114	-0.0886	481.731	573.277	False
10	3188	122.408	33.7355	3.6285	0.9613	0.8804	-0.0452	529.624	391.03	False
11	3013	170.645	29.6128	5.7626	0.9848	0.5248	-0.2083	594.958	912.812	False
12	3414	148.999	33.1799	4.4906	0.9749	0.8294	-0.0805	580.581	782.628	False
13	4647	139.214	45.6902	3.0469	0.9446	0.8623	-0.0076	723.033	247.267	False
14	1123	92.9757	18.8182	4.9407	0.9793	0.6997	-0.1287	817.288	724.433	False
15	3750	145.258	33.093	4.3894	0.9737	0.8965	-0.0485	952.204	400.276	True
16	1341	90.1135	19.5596	4.6071	0.9762	0.911	-0.0956	984.752	948.262	True

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	5	203	553	677.286	22	30.7857	6.9395
2	4	187	492	781.742	30	26.0581	7.3126
3	8	288	696	1008.62	41	24.6005	6.256
4	7	231	634	457.647	17	26.9204	6.4503
5	9	334	975	2532.69	106	23.8933	6.531
6	18	743	1607	4800.7	199	24.1241	6.704
7	9	383	835	983.867	51	19.2915	6.6928
8	16	431	863	1726.14	79	21.8498	6.7312
9	7	391	1003	3049.38	118	25.8422	7.7634
10	12	435	1212	2345.32	90	26.0591	6.864
11	16	678	1693	3915.52	186	21.0512	6.7003
12	16	536	1242	1773.18	77	23.0283	6.0764

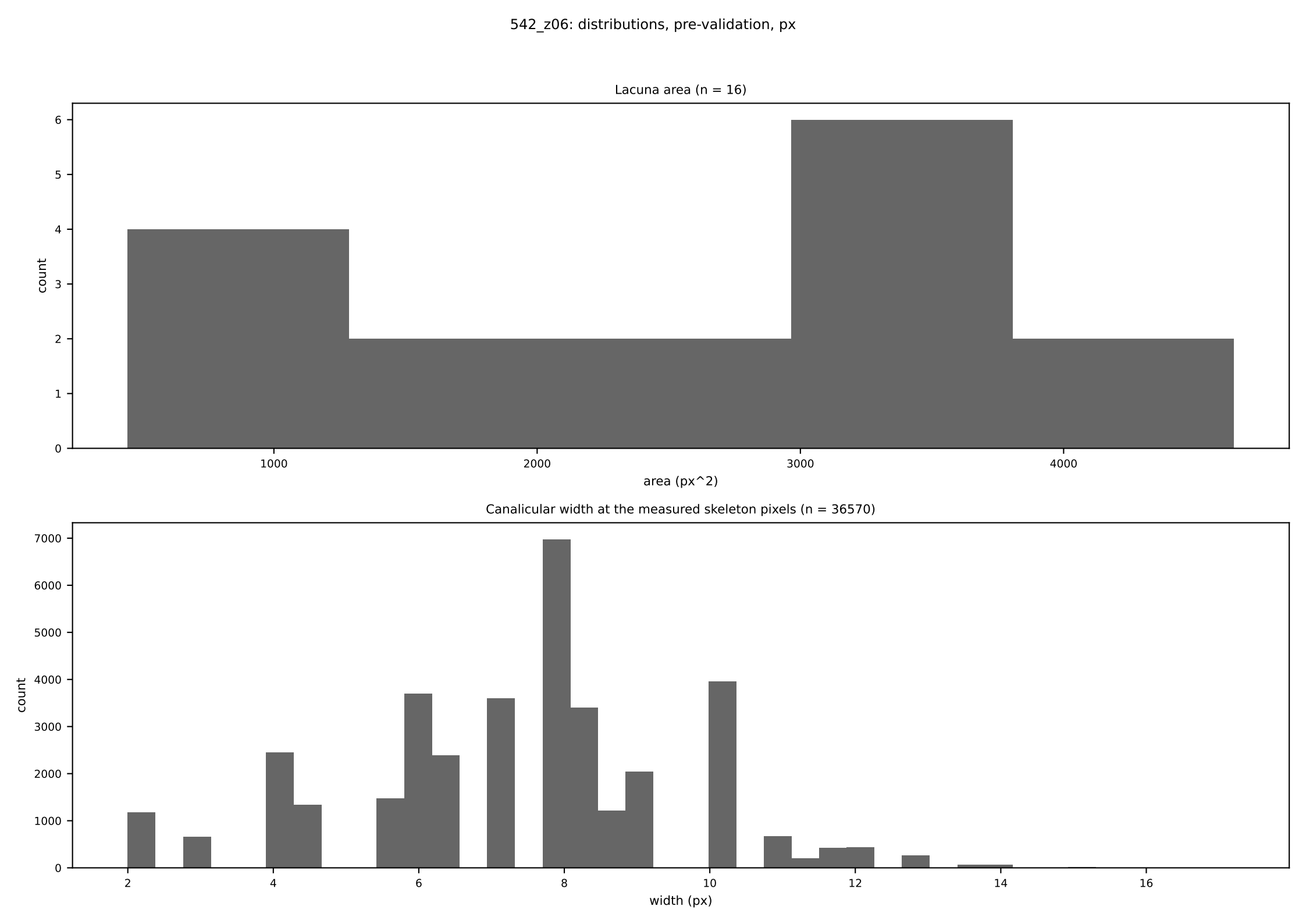
13	13	575	1251	2728.86	112	24.3648	6.6186
14	7	275	789	1305.15	40	32.6288	6.1517
15	4	261	668	839.441	40	20.986	6.7114
16	4	138	489	107.326	8	13.4157	4.7971

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	215.232	5617	12694	0.6363	0.5983	0.0361403	0.0435639	2.3231
2	126.468	6226	14794	1	1	0.0300353	0.0332567	3.1629
3	160.64	6976	18799	1	0.9774	0.0412844	0.0370233	4.9801
4	151.782	6752	15321	1	1	0.0342121	0.0413811	4.6119
5	322.043	11663	28685	1	1	0.0286376	0.0339899	2.7947
6	438.517	13901	31616	1	1	0.0534494	0.0508287	4.1047
7	211.924	8845	21455	1	1	0.0433013	0.0389187	4.2468
8	359.191	12383	27034	1	0.9025	0.0348058	0.0319228	4.4545
9	302.451	11295	27170	1	1	0.0346171	0.0369157	2.3144
10	309.522	11337	28286	1	1	0.0383699	0.042848	3.8769
11	468.274	15363	34096	1	1	0.044132	0.0496539	3.4168
12	422.978	13165	30001	1	1	0.040714	0.0413986	3.7827
13	379.848	13296	32108	1	1	0.0432461	0.0389622	3.4224
14	217.924	8830	22986	1	1	0.0311438	0.0343252	3.2121
15	332.894	10505	23806	0.8602	0.7919	0.0248453	0.0280601	1.2016
16	205.776	6766	16357	0.7841	0.714	0.0203961	0.0298955	1.9439

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	203	495	231.409	633.772	6	5	5
2	152	372	217.137	553.446	4	7	10
3	217	374	334.463	793.482	6	9	5
4	179	243	259.723	713.914	5	7	8
5	275	682	378.12	1100.96	7	12	13
6	713	1487	862.051	1863.18	18	18	23
7	312	664	427.706	941.352	8	12	13
8	385	668	478.09	964.767	9	12	10
9	313	555	441.161	1124.72	11	14	13
10	413	973	499.274	1389.7	10	14	15
11	600	1465	771.269	1904.32	14	25	24
12	404	839	591.446	1381.84	13	14	15
13	511	997	657.5	1425.01	13	21	20
14	160	377	306.451	890.453	5	10	14
15	178	271	287.966	739.428	6	9	13
16	103	103	152.711	540.304	4	6	6


542_z06: distributions, pre-validation, px
Lacuna area (n = 16)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 36570)
count
width (px)